

CURRICULUM VITAE

PROFESSIONAL SUMMARY

INDUSTRIAL RESEARCH & DEVELOPMENT (R&D) | SOFTWARE ENGINEERING & ML/DL:

- Dr.-Arch. & Senior Software Engineer with 9+ years of industry experience in professional software development, specializing in physical system simulation (EnergyPlus) and data-driven modeling.
 - Combines production-proven software architecture (Python, MATLAB) with advanced Machine Learning techniques (Scikit-Learn, TensorFlow, Docker) to engineer market-ready product solutions.
 - Demonstrated success in automating complex simulation pipelines, algorithmic optimization, and translating physical models into commercial software.
 - Target Roles: Senior Software Engineer (R&D), Python & Simulation Engineer, Machine Learning / AI Engineer.
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ADVANCED TRAINING

March 2026 – August 2026 | Certificate completed

- **Specialization: Software Development, Machine Learning & MLOps (760 Academic Units / 570 Hours)**
 - **Python 3 & Advanced OOP:** Modular development, data structures, unit testing, software architecture.
 - **Machine Learning & Deep Learning:** Scikit-Learn, TensorFlow, Keras, CNN/RNN, Ensemble Models.
 - **MLOps & Containerization:** Docker, Kubernetes, CI/CD pipelines, local ML deployment.
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PROFESSIONAL EXPERIENCE

January 2018 – January 2026 | Building Energy Efficiency Consultant & Software Developer

METABUILD GmbH, Berlin, Germany

- **Software Engineering & Enhancements:** Code development (MATLAB, SQL, Tera/Jinja2) and optimization for indoor air quality and thermal comfort metrics.
- **Scientific Integration:** Implemented mathematical and physical algorithms directly into production software.
- **System Simulation:** Modeled and analyzed building energy and HVAC system logic within simulation engines (EnergyPlus) to optimize overall system performance.
- **Feature Integration:** Advised on and developed core building performance features within company software systems.
- **Building Energy Modeling (BEM):** Implemented BEM workflows derived from BIM modeling platforms (IFC files, Revit).
- **Engineering Consulting:** Specialist advice on materials, structural engineering, building typologies, and retrofits.

2016 – 2018 | Building Energy Efficiency Consultant & Technical Support Specialist

ECOGLOBE GmbH, Berlin, Germany

- **Technical Support & Solutions:** Led debugging, qualification, and execution of custom solutions for complex client requirements.
- **Software Development:** Engineered code (MATLAB, LaTeX, Lua) to enhance daylighting algorithms using EnergyPlus and post-processing script workflows in MATLAB and reports.
- **Core Software Enhancements:** Integrated scientific calculations and numerical models into software systems.
- **Energy Modeling:** Developed BEM models for daylight calculations and automated data processing pipelines.
- **Cross-Functional Collaboration:** Worked closely with core software development teams to specify product features and improve UX/UI usability.

2006 – 2015 | STRUCTURAL & ENGINEERING RESEARCH (Selection)

- **2011–2015 Researcher** (Technical University of Madrid & Complutense University of Madrid, Campus-Moncloa & UdK Berlin): Focused on building energy efficiency and heat transfer dynamics. Utilized complex simulation tools (EnergyPlus, Ecotect, Simulink, MATLAB) and fast prototyping via 3D printing.
- **2013–2015 Assistant Professor** (Technical University of Madrid - U.P.M.): Taught advanced construction and architectural engineering techniques; constructed scale models for scientific experimentation.
- **2006–2009 Project Architect** (AMBITO-eme, Madrid): Responsible for all project lifecycle phases (HOAI 1–9 equivalent), including cost estimation and structural engineering for residential buildings.

2001 – 2006 | Architectural Intern / Trainee

Various Architectural Practices, Madrid, Spain

- Gained hands-on experience in architectural engineering design, drafting, and project management assistance

EDUCATION – Technical University of Madrid (U.P.M)

- **Ph.D. in Architecture (Dr. Arch.)** – International Doctorate Mention | 2010 – 2016
 - Dissertation: "Simulation of Thermal Performance of Glazing in Architecture using Scale Models"
- **Master of Science (M.Sc.)** in Building Structures | 2009 – 2010.
- **Degree in Architecture (B.Arch & M.Sc. Arch)** | 1994 – 2006

TECHNICAL SKILLS

Category	Skills/Knowledge
DevOps, OS	Docker, Kubernetes, Git, GitHub, Windows/Linux, NixOS, VIM, Microsoft Office, Mattermost
Machine Learning, AI	Tensorflow (Advanced), Keras (Advanced), Scikit-Learn (Advanced)
Programming Languages	Python/OOP (Advanced), Matlab (Advanced), Tera/Jinja2 (Advanced), LaTeX (Advanced), Lua (Intermediate), SQL (Intermediate)
BIM & CAD Software	AutoCAD, Revit, SketchUp
Specialized Software Engines (BEM)	Simulation: EnergyPlus, DesignBuilder, Metabuild Studio
Other Software	ArchiCAD, CYPE Structural, SAP Structural, 3D Studio Max, CorelDraw, Photoshop

LANGUAGES:

Spanish: Native | **German:** C1 | **English:** C1

SELECTED PROJECTS, PATENTS & PUBLICATIONS

- **Patents:** Co-inventor of software control and energy management systems.
- **Publications:** Author of multiple peer-reviewed scientific papers on energy efficiency and thermal dynamics in buildings. First author of primary research published in the high-impact journal Energy and Buildings (Elsevier): <https://www.sciencedirect.com/science/article/abs/pii/S0378778817306928?via%3Dihub>
- **Awards:** 2nd Place, Tech-Based Business Startup Competition – ACTUA-UPM (Focus: Renewable Energy Integration in Architecture).